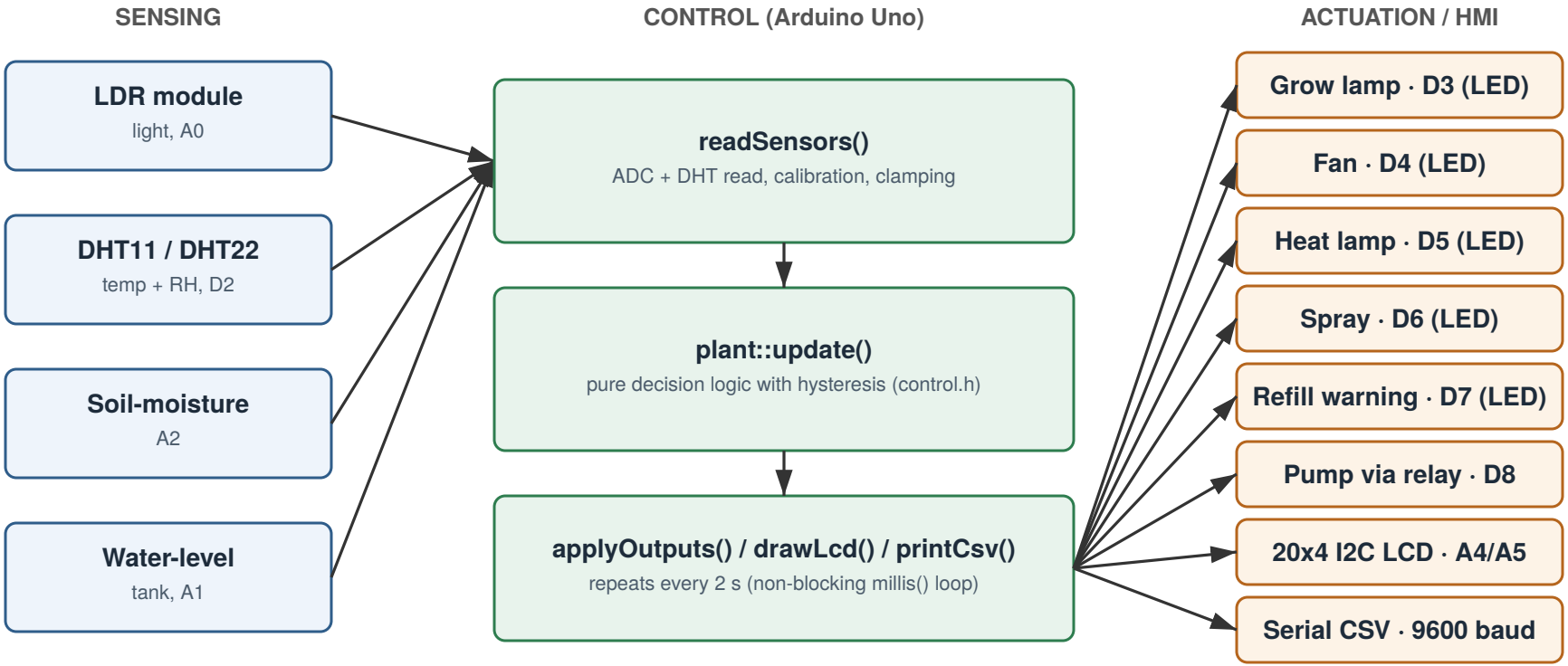
SENSING
CONTROL (Arduino Uno)
ACTUATION / HMI
LDR module
light, A0
DHT11 / DHT22
temp + RH, D2
Soil-moisture
A2
Water-level
tank, A1
readSensors()
ADC + DHT read, calibration, clamping
plant::update()
pure decision logic with hysteresis (control.h)
applyOutputs() / drawLcd() / printCsv()
repeats every 2 s (non-blocking millis() loop)
Grow lamp · D3 (LED)
Fan · D4 (LED)
Heat lamp · D5 (LED)
Spray · D6 (LED)
Refill warning · D7 (LED)
Pump via relay · D8
20x4 I2C LCD · A4/A5
Serial CSV · 9600 baud